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The rise and fall of K-Fashion business: exploring the impact of corporate failure news on stock market indices

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Abstract

Global demand for K-Fashion has paralleled the rise and fall of numerous South Korean firms. Drawing on the corporate life cycle and news priming theories, this study explores how online news on insolvent K-Fashion firms shapes the stock market's business cycle. In Study 1, we conducted dynamic topic modeling to analyze 35,960 news articles from 2010 to 2022. The findings revealed three categories of chronological topic changes, including sales performance improvement strategies, organizational restructuring, and the impact of global fashion labels. In Study 2, we employed multiple regression analysis to measure how insolvent K-Fashion firms' news and macroeconomic factors affect the clothing firms in the Korea Composite Stock Price Index. The results demonstrated that equity disclosure, bankruptcy, Samsung C&T's governance changes, and business diversification significantly influenced the stock market index. Integrating both natural language processing and structured data in this study offers practical insights to fashion firms and stock market investors.

Keywords: Fashion bankruptcy, Stock market prediction, Corporate bankruptcy, Business insolvency, Corporate life cycle, News priming

Introduction

South Korea is a leading spender in the global fashion industry (Lee, 2023), driven by its distinctive culture, high social media penetration, and influential personalities (Chitrakorn, 2023). The K-Fashion market was valued at approximately $24.4 billion (Statista, 2023), with a few key domestic fashion retailers achieving local and global recognition. Nevertheless, the sales of the majority of local fashion retailers have declined since the start of the 2010s (Ryu, 2011) owing to multiple factors, including subdued economic growth, rising real estate prices, and volatile monetary conditions (Kang, 2015; Korea Federation of SMEs, 2016; Samsung Global Research, 2012). The domestic fashion market gradually decreased, with 72% of new fashion labels being overseas licensing brands (Min, 2017) and clothing imports doubling in the past decade (Jo, 2023).

Corporate life cycle (CLC) theory, from organizational science, covers a firm's progression from emergence to decline, including the stages of introduction, growth and establishment, shake-out, and decline (Hasan & Habib, 2017; Zhang et al., 2021). The

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K-Fashion industry grew in the 1960s and 1970s with advancements in fiber and garment production. During this period, ready-to-wear and local ventures emerged to overcome the restrictions imposed by the Multi Fibre Arrangement (MFA), which set export quotas to protect industries in developed regions (Hale & Burns, 2005; Yu, 2021). Growth continued in the 1990s with global sourcing and local brands but soon faced challenges from International Monetary Fund (IMF) intervention, global competition, and financial crises (Kar, 2018). From 2010 to 2019, major firms disappeared due to restructuring and socioeconomic challenges, with the COVID-19 pandemic adding further disruption (Yoo & Ahn, 2022).

Various social and economic events have significantly affected the development of the K-Fashion industry. Between 2010 and 2015, several large local firms disappeared, resulting in governance restructuring, delisting, bankruptcy, and court receivership (Han, 2015; Jo, 2011; Ryu, 2011; Seoul Institute, 2014). From 2016 to 2019, the industry faced socioeconomic challenges, such as declining consumer demand, rising real estate prices, stagnant economic growth, and biflation—simultaneous inflation and deflation (Hong & Oh, 2018; KPMG, 2018). Moreover, the global outbreak of COVID-19 in 2019 led to lockdowns that severely disrupted fashion manufacturing and retail systems (Yoo & Ahn, 2022). However, scarce research investigates the trend changes during the decline stage of the K-Fashion industry or its correlation with business cycles like stock market fluctuations under economic conditions.

Media priming effects pertain to an individual's cognitive familiarity linked with information conveyed through the media (Hoewe, 2020). Notably, online news on economic and business firms significantly influences investors' perceptions, consequently impacting the stock market (Moy et al., 2016). Scholars have investigated the relationship between finance-related news and stock index fluctuations, focusing on fake news and market reactions (Clarke et al., 2020), investor sentiment analysis (Hanna et al., 2020), financial news and stock movement prediction (Nam & Seong, 2019), and stock forecasting based on media news (Shynkevich et al., 2016). However, the impact of online news about insolvent K-Fashion firms on the stock market amid macroeconomic environments remains unclear.

To bridge this gap in the body of knowledge, this study explores how online news on insolvent K-Fashion firms shapes the stock market's business cycle. We conducted two studies. Study 1 applied dynamic topic modeling (DTM) to analyze 35,960 news articles from 2010 to 2022, using the decline stage of the CLC theory to illuminate patterns in media coverage and their impact on the market. In Study 2, we employed multiple regression analysis to measure the effects of insolvent K-Fashion firms' news on clothing firms in the Korea Composite Stock Price Index (KOSPI), considering macroeconomic factors.[1] Priming effects demonstrate how consistent exposure to K-Fashion firm insolvencies impacts the fashion business cycles as reflected in stock indices. Through this two-fold analysis, the study aims to provide valuable insights into which issues concerning defaulting firms influence the overall performance of the fashion industry.

[1] The Korea Composite Stock Price Index (KOSPI) is used to represent the general trend of stock prices across various industries. In this study, the Textile & Wearing Apparel KOSPI index is used to represent the fashion industry's business cycle.

By combining DTM, text mining, and structured KOSPI data, this advanced approach offers a comprehensive analysis of media topics related to insolvent K-Fashion firms and their effect on the industry's performance.

Literature Review

The current stage of the K-Fashion industry in the corporate Life cycle

Corporate Life Cycle (CLC) theory describes the developmental stages of a corporation, from inception through growth, maturity, and eventual decline, each characterized by unique challenges and strategies (Zhang et al., 2021). Scholars in business management and finance refer to these stages in different ways, including introduction, growth, maturity, shake-out, and decline (Dickinson, 2011; Hasan & Habib, 2017) or as budding, growth, maturity, and recession (Zhang et al., 2021). In the introduction stage, business founders nurture new ventures despite limited capital (Faff et al., 2016). In the growth and establishment stage, as companies begin to generate consistent income, their market position strengthens, allowing them to expand their products and services (Wen et al., 2022). The culmination of this progression is the maturity stage, characterized by stable sales and profits as well as the efficient allocation of resources, capacity for technological innovation, and establishment of hierarchical organizational structures (Kishore et al., 2004). However, when the shake-out stage begins, companies face intense competition and threats. This stage triggers a transitional period in which they must determine whether to embrace innovation to survive or adopt alternative strategies such as mergers and acquisitions, delisting from the stock market, or restructuring (Hasan & Habib, 2017). Finally, in the decline stage, companies choose between a market exit through divestiture or a process of reinvention to rejuvenate their operations (Filatotchev et al., 2006).

We can apply CLC theory to South Korea's fashion industry. This industry transformed in the early 1960s and 1970s propelled by advancements in the garment sewing sector and the incorporation of synthetic fibers such as nylon and polyester, along with the government's economic development initiatives (Yu, 2021). The early 1970s to late 1980s was characterized by a strategic shift toward the burgeoning ready-to-wear market in response to the quotas of the MFA, redirecting the sector from export reliance to developing the domestic market (Kar, 2018). The mature stage during the 1990s was driven by mass production, international outsourcing, and the development of an intricate logistics system within K-Fashion firms after market liberalization (Ryu, 2011).

The competitive dynamics in the K-Fashion industry escalated throughout the 2000s. This came as a result of IMF intervention, rapidly expanding global fashion brands including luxury and fast fashion, and the global financial crisis (Ryu, 2011; Yu, 2021). As a result, some major K-Fashion firms experienced decline throughout the 2010s (e.g., TOMBOY, SSAMZIE, and ORANGE FACTORY), while others thrived by changing strategy, leveraging emerging technologies and undergoing organizational restructuring. Furthermore, the COVID-19 pandemic disrupted manufacturing in the K-Fashion industry, causing a decline in retail sales (Cho & Chyung, 2020). Thus, the decline stage of the K-Fashion industry has been intensified by external factors, including media priming effects that underscore competitive challenges, broader socioeconomic conditions, IMF-related economic pressures, the COVID-19 pandemic, and evolving consumer preferences.

Recent empirical studies in business management and finance that have explored insolvent firms based on CLC theory have focused on specific stages. Habib and Hasan (2019) classified CLC studies into those focusing on financial reporting, policies, and corporate governance, while Atif et al. (2022) examined the extent to which Environmental, Social and Governance (ESG) factors impact cash holdings across the life cycles of S&P 1500-indexed firms. Zhang et al. (2021), through the lens of CLC theory, investigated technological innovation as a crucial mediator between coagglomeration and environmental pollution in China's provinces.

Among studies on the fashion industry, Kozlowski et al. (2012) used life cycle assessment to examine the environmental impact of the fashion sector, while Fink (2003) elucidated the competitive strategies pursued throughout the life cycle. In South Korea, Yu (2021) examined the development of K-Fashion companies using CLC theory. However, their study centered on chaebols and lacked a media news analysis using large datasets. Hence, to the best of our knowledge, there has been no prior examination of the insolvency of local enterprises in the K-Fashion sector. This gap is particularly significant given that the decline stage of the CLC theory is marked by financial instability, organizational restructuring, and market exits, which are evident in the experiences of insolvent K-fashion enterprises. To address this gap, our research investigates insolvent K-fashion enterprises in their declining stages from 2010 to 2022, while comparing them to the prevailing socioeconomic conditions.

Socioeconomic and fashion industry trends in South Korea, 2010–2022

Socioeconomic trends refer to the broader patterns and changes in the social and economic factors that can influence the country's society over time. We compared South Korea's broad spectrum of socioeconomic trends with the changes in the fashion industry.

During the early 2010s, South Korea's use of smart technologies, including tablets, smartphones, and apps, rapidly evolved (Im et al., 2010). Concurrently, fashion retailers integrated information technologies within their stores. However, the rate of development declined markedly from 2011, leading to rising product prices alongside diminishing trends in consumption, employment, and real estate prices (Jo, 2011). From this point, consumers prioritized cost-effective products, while firms sought to increase sales through improved social media strategies (Lee, 2011).

In 2012, the Korean economy was racked by low growth and high prices (Samsung Global Research, 2012), and biflation followed in 2013 and 2014 (Chosun Biz, 2013). K-Fashion enterprises endeavored to expand internationally by leveraging the popularity of the *Hallyu* culture. Simultaneously, competition in the domestic market rose, primarily driven by the presence of global fast fashion brands and the diversification of fashion trends among consumers (Ko, 2012), while the boundaries between online and offline retail blurred (Kang, 2012).

During the mid-2010s, there were economic stagnation (distinguished by four consecutive quarters of a low growth rate in 2016) and a rapid decline in the appeal of local outdoor brands (Chosun Biz, 2016; Han, 2015; Kang, 2015). In response, many fashion firms engaged in mergers and acquisitions to avoid insolvency and diversified their business sectors through sales-driven strategies (Fashion Insight, 2016).

In the late 2010s, the economic uncertainty from Brexit and swift regulatory changes affected business investment, while the Fourth Industrial Revolution gained traction, accentuated President Jae-in Moon's election in 2017 (Korea Federation of SMEs, 2016). Furthermore, Samsung Electronics merged with Cheil Industries, Inc., a Samsung Group subsidiary (KPMG, 2018). In 2019, the structural downturn persisted, consumption patterns remained disparate, and the U.S. economic revival continued to have local repercussions. Simultaneously, the K-Fashion industry received substantial capital from fund companies and corporate strategic investors (Fashion Hong & Oh, 2018; Insight, 2019).

During the early 2020s, the global COVID-19 pandemic surfaced, rapidly shifting fashion retailers toward non-face-to-face environments (Statistics Korea, 2020). Moreover, the disruption of the global supply chain and the significant increase in real estate prices influenced the K-Fashion sector, prompting the implementation of strategies to increase sales (Federation of Korean Industries, 2021). When President Seok-yeol Yoon was elected in 2022, the economic environment was still characterized by persistent low growth, including year-round high inflation and escalating interest rates (Yoo & Ahn, 2022). The K-Fashion industry is gradually recovering from the COVID-19 pandemic and increasingly adopting digital technologies such as virtual reality and platforms driven by big data (Han, 2023). Considering the aforementioned insolvency issues in the K-Fashion industry and its relation with socioeconomic conditions, this study proposes the following research questions:

RQ1: How did the topics and keywords within media coverage of insolvent K-Fashion companies shift from 2010 to 2022?

RQ2: How are the topics of insolvent K-Fashion firms connected to socioeconomic trends?

Stock market response to the effects of media priming

The media priming effect refers to how the media influences individuals' responses to subsequent information (Hoewe, 2020). Media coverage is particularly important because it not only informs but also influences public opinion, investment behaviors, and market dynamics (Nam & Seong, 2019). In the context of our research, ongoing media coverage about struggling K-Fashion firms is crucial as it can amplify the perceived risks of insolvency, sway investor confidence, and affect stock market responses. When the media highlights specific issues, the spotlighted information is stored in one's memory, making it more accessible when evaluating new information. Hence, it pertains to an individual's cognitive familiarity linked with information conveyed through the media (Moy et al., 2016). Individuals retain information in their memory by creating nodes that correspond to similar concepts, and these nodes subsequently propagate to other stimuli. Specifically, media priming affects their audience's behavior and judgment through various modes (Bryant & Oliver, 2009; Piselli et al., 2022). The influence of media news has risen since the explosion in online news, strongly affecting the public's societal awareness and interpretation of a topic. Priming is closely related to agenda-setting theory, which explains how specific issues stand out more prominently in individuals' thoughts (Perloff, 2022). The primary focus of agenda-setting research has been on how mass communication influences the public's perception of an issue of importance (Chung et al., 2023). However, priming differs in that it can occur over a period of time following exposure to a particular media segment. Therefore, the priming effect clarifies that the mass media holds the power to shape the factors that individuals consider when addressing other matters (Moy et al., 2016). Hence, our study utilized the priming effect framework to examine how ongoing media coverage of struggling K-Fashion firms may influence the stock market over the following decade.

Price and Tewksbury (1997) emphasized that effective priming requires both applicability (where the highlighted concept from the media aligns with the target stimulus features) and subjective relevance (signifying a personal connection to the notion). Certain issues become primed in the public's perception when the media consistently releases similar topics, thereby influencing their judgments of the media messages (Pennycook & Rand, 2022; Van Duyn & Collier, 2019). According to Moy et al. (2016), individuals connect the information they receive from media channels (such as online news, TV, and magazines) with their existing knowledge. The media priming effect happens when people apply the concept highlighted by the media to a situation where they normally wouldn't (Hoewe, 2020). For example, media priming can influence stock market investors' decisions (Kliger & Kudryavtsev, 2010), the public's evaluations of the government (Kalogeropoulos et al., 2017), returns and volatility in oil markets (Mensi et al., 2014), and consumer behavior (Nguyen & Claus, 2013).

In particular, news about economic and business firms significantly impacts the stock market, reflecting industry health and business cycle stages. For example, positive news about fashion firms boosts investor confidence and raises the Textile & Wearing Apparel KOSPI prices, while negative news causes concern and lowers stock prices. Many scholars have discussed this interrelation. Nam and Seong (2019) adopted a machine learning approach to examine stock price movements using financial news and established a causal relationship. Additionally, Kim and Park (2023) analyzed the correlation between South Korea's fashion industry news and clothing companies' stock market indices using autoregressive integrated moving average models with exogenous variables, based on framing theory, which is a subsequent framework of media priming effects. Furthermore, previous studies (Clarke et al., 2020; Groening & Kanuri, 2018; Hanna et al., 2020; Lee & Brahmasrene, 2018) have shown that news priming provides compelling evidence of a correlation between corporate news and the business cycle. Thus, we speculate that news on the insolvency of K-Fashion firms could affect clothing firms in the KOSPI in conjunction with the macroeconomic circumstances.

Strategic approach to integrating DTM and structured data

Topic modeling refers to using a category of unsupervised machine learning algorithms to uncover the latent themes within lengthy texts. These algorithms extract multiple topics, each formed from unique combinations of words that tend to appear together frequently across the documents in a corpus (Nielsen & Borjeson, 2019). Latent Dirichlet allocation (LDA) is commonly employed in text mining because it can estimate both the distribution of words by topic and the distribution of topics by document, thereby enabling researchers to identify the topics present in each document (Yin & Yuan, 2022). However, as LDA assumes uniform priors for both topics and words across all documents, it cannot capture temporal changes. As Müller-Hansen et al. (2021) explained, DTM relies on a generative statistical model tailored to the data, assuming that the weights of the terms within topics and topics within documents are drawn from Dirichlet distributions. Consequently, we use DTM to monitor the consistent fluctuations of topics by tracking the changes in prominent topics and their associated keywords over time.

Benefiting from the rapid advancements in big data analysis, a growing number of scholars are adopting integrated methodologies that leverage both text mining and structured data analysis. For instance, Burggraf et al. (2020) examined the extent to which news on economic fluctuations affects the KOSPI 200, and BL and BR (2023) predicted stock prices by adopting news sentiment analysis. Yu et al. (2023) revealed the positive impact of sentiment extracted from ESG-related news in reducing stock price crashes. However, studies on the fashion industry have relied on limited methodologies, including using qualitative methods to examine the life cycles of fashion firms (Yu, 2021), time series analysis to investigate the evolution of fashion industry news topics (Kim & Park, 2023), and machine learning techniques to predict clothing sales (Lv et al., 2023). Thus, it is necessary to explore the specific insolvency news on K-Fashion firms that influence the fashion business cycle within financial market analysis. Thus, it is necessary to explore the specific dimensions of K-Fashion firms' insolvency news that influence the business cycle within the realm of financial market analysis. Accordingly, this study proposes the following research questions:

RQ3: What news topics on the insolvency of K-Fashion firms significantly change the stock market and macroeconomic indices?

RQ4: How are news topics on the insolvency of K-Fashion firms related to the stock market and macroeconomic indices?

Study 1

In study 1, we collected digital news on insolvent K-Fashion firms from the past 12 years. The data refining and cleansing process involved morphological analysis, stop words, term frequency (TF), and term frequency-inverse document frequency (TF-IDF). Interpretability and reliability tests confirmed the topic subjects and time series changes of the keywords through DTM representations. The prominent topics were then compared with economic and social issues to comprehend K-Fashion insolvency trends and simultaneous domestic changes, addressing RQ1 and RQ2.

Methods

Data collection

We collected the online news data from South Korea's largest online portal (Naver's media affiliates, https://news.naver.com/) and used several Python libraries. Specifically, the library "os" facilitated interactions with the operating system, enabling file operations and directory manipulation. Additionally, "datetime" and "timedelta" libraries were employed to set the dates and times. Then, HTTP requests and URL openings were executed using the "urllib.request" library and "lib_common" was incorporated to specify the news data crawling. We adopted the search keywords "reconstructing," "delisting," "disposal," "bankruptcy," "insolvency," and "court receivership" (Bessler et al., 2023; Levratto, 2013; Olujobi, 2021) in conjunction with "fashion." Through web crawling, 35,960 news articles were collected from 612 economic and business media sources between January 1, 2010, and July 31, 2022.

Refining and cleansing the data

To eliminate irrelevant and duplicated content, real estate auction advertising, and stock recommendations, articles without text were deleted, leaving 34,247 news articles. Then, we employed the Komoran package (Korean morphological analyzer) to extract Korean nouns from natural sentences because adjectives, adverbs, and verbs are represented effectively in sentiment analysis (Suh, 2019). The TF-IDF algorithm was examined to evaluate a term's significance in the overall corpus relative to its rarity across the collection, while TF measured the frequency of certain words within a document (Hussain et al., 2022). Subsequently, we examined word sequences using n-gram-based methods to establish the proper nouns (e.g., Ebay → Ebay Korea, Everland → Samsung Everland, control & structure → governance) and unify synonyms (e.g., apparel, clothing, clothes → clothing, global, overseas, international → global). This process involved the removal of stopwords, special characters, numbers, and emojis.

Measurement and research process

In DTM, it is assumed that every document encompasses various topics and that the words in those documents are generated from a multinomial distribution. First, we cleaned and tokenized the text data, creating a dictionary and converting the text into a bag-of-words format. Next, we examined the coherence scores (Bogdanowicz & Guan, 2022) to confirm the optimal number of topics and explored qualitative reviews based on the descending percentage of topic contribution and weight per news article using the LDA algorithm (Kim & Park, 2023). Intercoder reliability was assessed by two fashion business experts who hold doctorate degrees in clothing science and have ten years of experience in the fashion industry. Then, Gensim, the Python library for natural language processing, was used to apply the DTM (Ha et al., 2017). We segmented the data into time slices and trained models incrementally, reviewing topic evolution to extract and visualize changes over time. Lastly, we performed a time series analysis to examine the annual changes in topic proportions, enabling us to compare South Korea's prominent socioeconomic issues with the dominant topic each year (Fig. 1).

Study 1 results

TF and TF-IDF analysis results

Table 1 shows the top 30 of the 4182 terms from the TF and TF-IDF analyses. The top 10 highest terms in the TF results were "brand," "acquisition," "share," "market," "Samsung Group," "invest," "China," "representative," "chairman," and "performance." The top 10 highest terms in the TF-IDF analysis were "brand," "acquisition," "Samsung Group," "share," "China," "market," "invest," "chairman," "representative," and "Samsung C&T."

Note that the most frequently found keyword ("brand") was the same in both analyses. This indicated that the companies' growth strategies centered on the fashion brand. Further, market share appeared mostly in the TF analysis, whereas information about

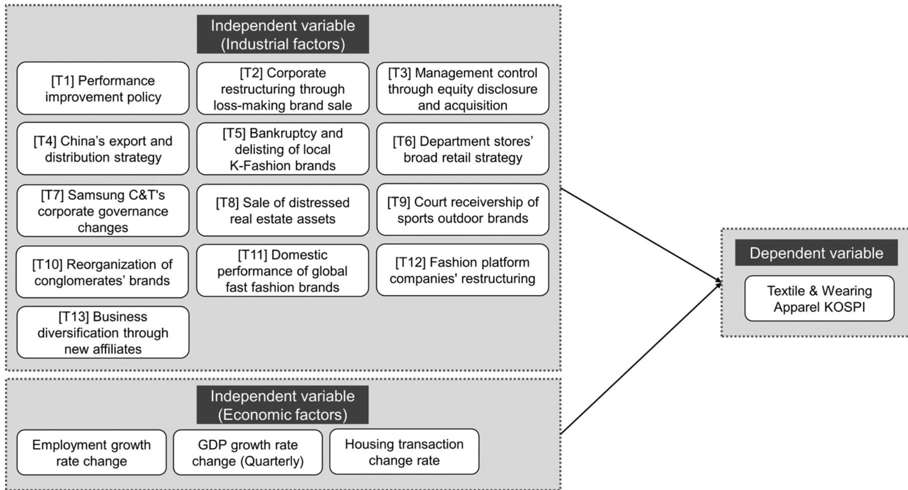


Fig. 1 The research model of the Study 2. [T1]–[T13]: Topics 1 to 13

Table 1 Results of the TF and TF-IDF (N = 4182)

Rank	Word	TF	Word	TF-IDF	Rank	Word	TF	Word	TF-IDF
1	Brand	8940	Brand	14,614.94	16	Distribution	2917	Cheil Industries Inc	7889.26
2	Acquisition	5387	Acquisition	11,305.98	17	Industry	2884	Industry	7870.18
3	Share	4388	Samsung Group	10,545.61	18	Growth	2821	Distribution	7538.29
4	Market	4142	Share	9868.88	19	Cheil Industries Inc	2776	Sales	7454.52
5	Samsung Group	3884	China	9646.18	20	Sales	2754	Growth	7393.60
6	Invest	3874	Market	9440.13	21	Clothes	2454	SK Networks Co. Ltd	7012.59
7	China	3785	Invest	9115.91	22	SK Networks Co., Ltd	2420	CEO	6995.12
8	Representative	3524	Chairman	8878.26	23	CEO	2266	Clothes	6739.74
9	Chairman	3435	Representative	8805.20	24	Danger	2258	Structure	6715.48
10	Performance	3419	Samsung C&T	8458.72	25	Shinsegae department store	2173	Danger	6542.39
11	Company	3358	Performance	8451.76	26	Profit	2021	Profit	6221.75
12	Improvement	3252	Company	8231.99	27	Overseas	1957	Shinsegae department store	5959.91
13	Sales	3173	Improvement	8122.10	28	Merger	1957	Affiliates	5903.35
14	Management	3147	Sales	8049.96	29	Asset	1945	Overseas	5863.05
15	Samsung C&T	2940	Management	7917.33	30	Revenue	1933	Propulsion	5765.35

TF: Term frequency

TF-IDF: Term frequency-inverse document frequency

major conglomerates' business investment and infrastructure was notable in the TF/IDF analysis.

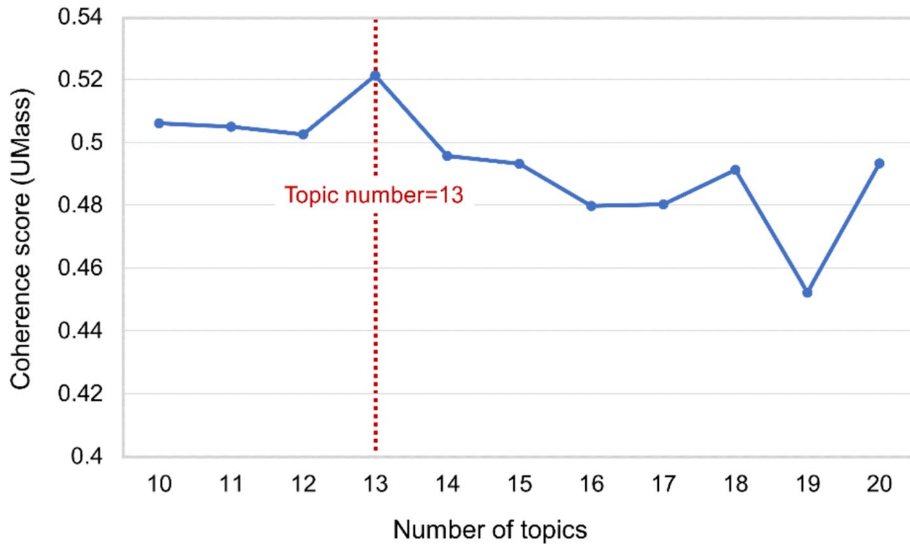


Fig. 2 Static LDA coherence scores for varied numbers of topics

Interpretability and reliability

According to Maier et al. (2021), it is important to preprocess both interpretability and reliability to evaluate the validity of unsupervised probabilistic topic models. We applied intrinsic coherence scores using UMass to measure the optimal number of topics and the extent to which each topic explained the variable topic count (Bogdanowicz & Guan, 2022; Tian & Fang, 2019). In Eq. (1), $D\ (w_i, w_j)$ is the number of times w_i and w_j appear together in the document, while $D\ (w_i)$ is the number of times w_i appears alone. The higher the number of UMass, the higher is the probability of w_i and w_j appearing together. This implies that the semantic relationship between words that frequently appear together in each topic is strongly associated. Consequently, it can be determined that our topic classification is consistent. Figure 2 shows that the coherence scores peak at topic 13, with a moderate decrease thereafter.

The UMass equation for determining the optimal number of topics

$$C_{UMass}\left(w_i, w_j\right) = \log\frac{D\left(w_i, w_j\right) + 1}{D(w_i)} \tag{1}$$

To verify the reliability of the topic content, the inter-rater reliability of the two field experts was assessed using Cohen's kappa, a statistical measure of intercoder reliability (Figueroa et al., 2023). Subsequently, we reviewed the news articles in ascending order of the topic index; multiplied the percentage topic contribution by the topic index per document; and used the LDA algorithm to display the article title, date, and content of the representative document with the highest topic index. The reliability among coders shows the consistency of the content. In this study, Cohen's kappa coefficient was 0.85, representing almost perfect agreement between the raters (Figueroa et al., 2023). Additionally, using the pyLDAvis package, an inter-topic distance map was developed to verify whether the topics were structured into identifiable subject clusters.

Implementing the DTM representations

Thirteen topics with keywords, defined as [T1]–[T13] below, were derived from the DTM algorithm. The five highest frequency keywords are based on the sum of the values used in pyLDAvis, an interactive visualization device designed to interpret the results of topic models. These 13 topics can be broadly categorized into the following three areas faced by insolvent K-Fashion enterprises over 2010–2022: sales performance improvement strategies (e.g., [T1], [T3], [T4], [T6], [T10], [T13]), corporate restructuring and financial challenges (e.g., [T2], [T5], [T7], [T8], [T9], [T12]), and the local market influence of global fashion brands (e.g., [T11]). Table 2 shows representative documents with a high topic index content and the inter-topic distance map.

The 13 topics were as follows. [T1] "Performance improvement policy" was related to fashion companies' efforts to increase profits by diversifying their business strategies into new retail channels and shopping entertainment mediums (Bae & Na, 2019; Jang, 2020a, 2020b; Kim, 2014, 2018). Among the top five keywords, "improvement" was initially ranked the highest before decreasing, while "sales," "performance," "turnover," and "profit" significantly increased over 2010–2022.

[T2] "Corporate restructuring by selling loss-making brands" was associated with fashion firms rebuilding by targeting underperforming local brands such as NII, KAI-AAKMANN, ADVISORY, and JACK AND JILL during the financial crisis (Jeon, 2019; Kim, 2016a, 2016b, 2021a, 2021b, 2021c, 2021d; Lee, 2018). This strategic change directly impacted exports in the textile industry. The prominence of the keywords "textile," "economy," and "industry" gradually declined over the study period, while that of "crisis" and "export" increased.

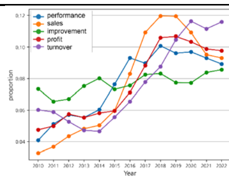
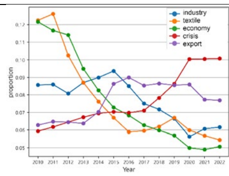
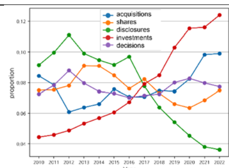
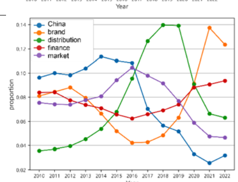
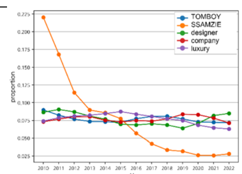
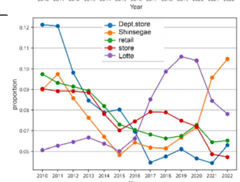
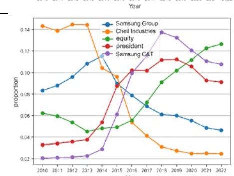
[T3] "Management control through equity disclosure and acquisition" represented changes to the management of fashion firms (An, 2018; Choi, 2013; Im, 2018; Oh, 2010). The importance of the main keyword "investment" gradually increased over the study period, while "disclosure" steadily decreased. The other keywords "shares," "acquisition," and "decisions" remained relatively stable.

[T4] "China's export and distribution strategy" implied the changes to trade policy on entering China's retail market due to international issues such as free trade agreements and political conventions (Jo, 2012; Lee, 2015a, 2015b, 2016; Oh, 2015). The main keywords "China" and "market" rose 2010–2016 and then declined, while "brand" and "distribution" moderately increased. Meanwhile, the importance of "finance" remained stable.

[T5] "Bankruptcy and delisting of local K-Fashion brands" was associated with the financial collapse of domestic K-Fashion brands (e.g., TOMBOY, DECO, SSAMZIE, and CORDEZ-COMBINE). This was caused by both internal concerns (e.g., management fraud, corporate governance battles, and falling sales) and external impacts (e.g., the popularity of global luxury fashion labels) (Choi, 2020; Kim, 2015a, 2015b; Kyungin Ilbo, 2010; Park, 2011). The importance of the main keyword "SSAMZIE" steadily decreased over the study period, while "TOMBOY," "designer," "company," and "luxury" remained stable.

[T6] "Department stores' broad retail strategies" related to major department stores such as Lotte, Hyundai, and Shinsegae diversifying their retail growth strategies by reintroducing TV home shopping, expanding subsidiaries, and renovating stores despite

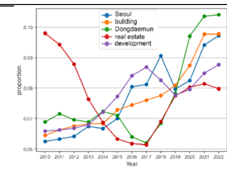
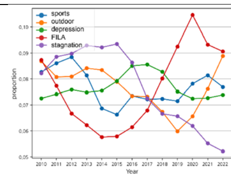
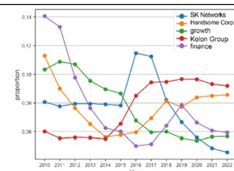
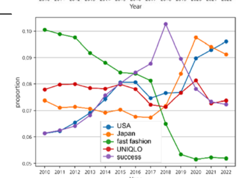
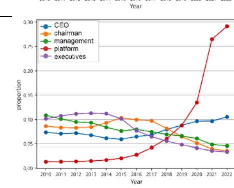
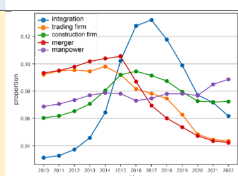
Table 2 Result of the dynamic topic modeling about K-Fashion insolvent company news (2010 January–2022 July)

Topic	Main keywords	Time series transition (Top 5 keywords)	Representative documents (news articles)	Topic per cont*weight
[T1] Performance improvement policy	performance sales improvement profit turnover increase invest effect prospect revenue		"Shinsegae International Inc. at the end of restructuring and strengthening online channels." (Jang, 2020)	14.44
			Comparison of competitiveness by distribution sector - department store, premium, 'Shoppertainment' differentiation...(Bae & Na, 2019)	13.53
			Kim Ki-san, Vice Chairman of the Fashion Association urgent improvement of unfair practices in large department stores and TV home shopping (Kim, 2014)	10.11
			Fashion industry overcomes low-listed limit by revitalizing brands (Kim, 2018)	6.07
[T2] Corporate restructuring through selling loss-making brands	industry textile economy crisis export government support factory technology production		"Young casual slump"… Sejeong Group sells casual brand "NII" (Kim, 2021)	9.56
			The casual market that collapses due to the suspension of "KAI-AAKMANN", "ADVISORY", "JACK AND JILL" (Lee, 2018)	5.91
			Hyungji-Elite Corp. failed to turn surplus again in clearing out bad inventory (Jeon, 2019)	4.86
			Restructuring the fashion industry... Speed of brand clearance/sales (Kim, 2016)	4.56
[T3] Management control through equity disclosure and acquisition	acquisitions shares disclosures investments decisions listing stock major shareholder KOSDAQ transaction		'Irresponsible owner family' leaving Dongyang Group (Choi, 2013)	8.26
			Why is the private loan company pleased with the management dispute? (Oh, 2010)	8.16
			Seong-soo Park overcomes the E-Land crisis with a professional management system (Im, 2018)	3.18
			Brands that have changed owners, life and death to recover earnings (An, 2018)	2.30
[T4] China's export and distribution strategy	China brand distribution finance market Eland Group beauty cosmetics TEENIE WEENIE entry		FTA will trigger rebound in exports next year (Oh, 2015)	9.88
			In the era of China's new normal, China should actively utilize FTAs to preempt domestic demand (Lee, 2015)	6.69
			Achievements such as Korea-China Conference on Joo Hyung-hwan, expansion of mutual investment, and elimination of non-tariff barriers (Lee, 2016)	5.85
			Recent changes in the retail distribution market in China (Jo, 2012)	5.73
[T5] Bankruptcy and delisting of local K-Fashion brands	TOMBOY SSAMZIE designer company luxury Italy France trend local New York		Fallen K-local fashion "TOMBOY" who will be the new owner? (Park, 2011)	7.36
			Women's clothing K-local fashion brand 'DECO' will be sold (Choi, 2020)	6.32
			"SSAMZIE" final bankruptcy... delisting (Kyungin Ilbo, 2010)	1.17
			Dongdaemun's legendary brand "CORDEZ-COMBINE" for sale (Kim, 2015)	.99
[T6] Department stores' broad retail strategy	department store Shinsegae retail store Lotte E-MART product large consumption discount store		"Lotte Shopping continues to lose 30 billion won in pre-tax profit due to burdensome debt" (Jang, 2020)	20.54
			Shinsegae Decreases 80% and E-Mart Surges 60%...One roof of operating profit, joy and sorrow (Maeng, 2021)	6.60
			The profits of the 3 distribution dinosaurs are growing... Worried about deteriorating cash flow (Cha, 2022)	5.92
			The splendid revival of home shopping... Booming alone amid economic stagnation (Ji, 2017)	4.25
[T7] Samsung C&T's corporate governance changes	Samsung Group Cheil Industries equity president Samsung C&T Everland Inc. subsidiary Seohyun Lee Jaeyong Lee Samsung Electronics		[Samsung's new management 23rd anniversary] Jae-yong Lee focuses on pragmatism under the banner of new Samsung (Park, 2016)	4.67
			[The direction of Samsung Group President Boo-jin Lee and Seo-hyun Lee] Holding a 5.1% stake in Samsung C&T (Park, 2016)	2.69
			Samsung C&T's Fashion Division, emergency management performance remaining tasks (Bang, 2021)	2.17
			Samsung's restructuring for the first time in 18 years...100 years of perpetual enterprise leap forward (Myung, 2014)	.82

economic stagnation (Cha, 2022; Jang, 2020a, 2020b; Ji, 2017; Maeng, 2021). The significance of the main keywords "Shinsegae" and "Lotte" rose in 2015 after decreasing in 2010. However, "department store," "retail," and "store" fell gradually over the 12-year period.

[T7] "Samsung C&T's corporate governance changes" represented governance reforms for Samsung C&T after mergers and acquisitions between its affiliates such as Chile Industries and Everland Inc. (Bang, 2021; Myung, 2014; Park, 2016a, 2016b).

Table 2 (continued)

Topic	Main keywords	Time series transition (Top 5 keywords)	Representative documents (news articles)	Topic per cont*weight
[T8] Sale of distressed real estate assets	Seoul building Dongdaemun real estate development estate hotel culture shopping mall bidding		Fashion building owners are branded as insolvent companies (Kbiznews, 2016)	4.30
			Sell or lose...Fashion companies are also warned of comprehensive real estate taxes (Lee, 2020)	3.95
			Dongdaemun Doosan Tower sold for 800 billion won (Kyung, 2020)	2.83
			Establishment of a real estate trust company in the financial sector – Shinhan Bank, Fashion LF Corp… 12 new companies also announced (Park, 2018)	2.62
[T9] Court receivership of sports outdoor brands	sports outdoor depression FILA stagnation court receivership LECAF Hwaseung Group corporate rehabilitation middle market enterprise		Falling sports, outdoor brands…Where is the best store? (Kim, 2019)	1.32
			Hwaseung Group, Korea's the first shoes company, applies for corporate rehabilitation (Lee, 2019)	.72
			"FILA" laughing, "LECAF" crying... The CEO's strategy is doomed (Im, 2019)	.33
			Sports brand "LECAF" enters rehabilitation process (Choi, 2019)	.19
[T10] Reorganization of conglomerates' brands	SK Networks Handsome Corp. growth Kolon Group finance improvement Duty-free shop Hyundai Department Store reorganization power		Hyundai Home Shopping & Handsome Corp., fantastic pairs…accompanied by the same goals (Kim, 2016)	1.20
			A busy year with SK Networks Co. Ltd.'s reorganization (Noh, 2017)	.99
			Close fashion and duty-free shops, add rental businesses...SK Networks Co., Ltd., after adjusting the business, is much better (Na, 2017)	.99
			Hyundai Department Store's Chairman, Jeong Ji-sun's magic... Handsome Corp.'s fashion wings (Lee, 2015)	.98
[T11] Domestic performance of global fast fashion brands	USA Japan fast fashion UNIQLO success bankruptcy ZARA FOREVER 21 online H&M		LS Networks Co. Ltd cries and laughs at Japanese brand (Beom, 2019)	7.93
			"UNIQLO" South Korea sales of 900 billion won... domestic brand cruelty (Oh, 2014)	5.82
			Up to 80% off "FOREVER 21" with store clearance (Byun, 2019)	5.07
			The image of "ZARA" has fallen... realization of the boycott movement (Kim, 2016)	4.24
[T12] Fashion platform companies' restructuring	CEO chairman management platform executives restructuring director Shinwon corp. vice president Kakao Corp.		Jung Yong-jin of action vs Shin Dong-bin of substance...a ten-year rival management battle (An, 2021)	7.51
			An online fashion platform company coveted by Shinsegae and Kakao... Excluding traditional clothing companies (Kim, 2021)	5.20
			Kakao Corp. acquires "ZIGZAG" instead of eBay...Now, Naver, Coupang, and Kakao are the best shopping malls in South Korea (Kim, 2021)	4.97
			Kakao Corp., fashion platform "ZIGZAG" acquisition of 1 trillion won value…What is the driving background? (Kim, 2021)	3.82
[T13] Business diversification through new affiliates	integration trading firm construction firm merger manpower launch synergy employee hope COVID-19		Kolon Group's bio sector is visible and its corporate value is highlighted (Kim, 2015)	2.70
			Doosan Group, infrastructure core and construction engineers are influential (Park et al., 2010)	2.24
			One of the secrets of COVID-19 related beneficiary companies is rapid business transformation (Lee, 2020)	.99
			Why is the fashion industry the first to undergo restructuring in two months due to COVID-19? (Bae, 2020)	.98

Note: 1) Cohen's Kappa value: .85
2) [T1]–[T13]: Topics 1 to 13
3) DTM categories

The keywords "equity," "Samsung C&T," and "president" rose consistently throughout the study period, whereas "Cheil Industry" and "Samsung Group" decreased after 2014.

[T8] "Sale of distressed real estate assets" comprised the selling of distressed real estate property (e.g., buildings, residential property, shopping centers, and malls) by fashion enterprises, whose brands are strongly associated with real estate asset transactions (Kbiznews, 2016; Kyung, 2020; Lee, 2020a, 2020b; Park, 2018). The importance of the main keywords "Seoul," "building," and "development" gradually increased from 2010–2022, while "real estate" and "Dongdaemun" dropped steadily until 2017 before increasing.

[T9] “Court receivership of sports outdoor brands” consisted of the prudent management of local K-Sports outdoor brands due to the market slowdown, rise in licensing fees, and intensifying global competition (Choi, 2019; Im, 2019; Kim, 2019; Lee, 2019). The keywords “sports,” “stagnation,” and “depression,” fell continuously over the study period, whereas “FILA” increased. Additionally, “outdoor” decreased steadily from 2010 to 2018 but increased from 2019.

[T10] “Reorganization of conglomerates’ brands” signified the consolidation of fashion firms (e.g., SK Networks, Handsome Corp., and Courronne) under major companies (e.g., Hyundai Department Store and Kolon Group) to maximize profits (Kim, 2016a, 2016b; Lee, 2015a, 2015b; Na, 2017; Noh, 2017). The significance of the primary keyword “Kolon Group” grew over the period, while that of “growth” decreased and “SK Networks” rose from 2015 to 2017, followed by a gradual decline. Moreover, “Handsome Corp.” and “finance” increased from 2016 until 2022 after a steady decline from 2010 to 2015.

[T11] “Domestic performance of global fast fashion brands” represented the impact on national sales of international fast fashion brands (e.g., FOREVER 21, UQNIQLO, and ZARA) (Beom, 2019; Byun, 2019; Kim, 2016a, 2016b; Oh, 2014). The importance of the main keyword “Japan” increased steeply from 2017 until 2020, while “fast fashion” fell. Additionally, “USA” gradually grew and “UNIQLO” maintained a stable position. However, “success” decreased from 2018, following a steady increase from 2010–2017.

[T12] “Fashion platform companies’ restructuring” indicated that online fashion platform companies (e.g., ZIGZAG and W-CONCEPT) merged into or were acquired by major firms (Kakao Corp. and Shinsegae Group), leading to a change in senior management (An, 2021; Kim, 2021a, 2021b, 2021c, 2021d). The popularity of the primary keyword “platform” steadily increased and then surged from 2018 to 2022, while “management,” “chairman,” “CEO,” and “executives” remained at modest levels.

[T13] “Business diversification through new affiliates” suggested that firms diversify their businesses by forming new affiliates, including firms in trading, construction, and engineering as well as in the fashion sector. This shift is influenced by factors such as COVID-19, concerns about future business growth, and development opportunities (Bae, 2020; Kim, 2015a, 2015b; Lee, 2020a, 2020b; Park et al., 2010). The main keyword “integration” steadily increased from 2010 to 2017 and then decreased until 2022, while “trading firm” and “merger” declined steadily. Additionally, “construction” gradually increased, then dropped from 2016, and “labor” maintained a stable position.

A comparative time series analysis of South Korea’s socioeconomic issues and DTM topics In the next step, we investigated the time series changes in the topic percentage to identify the most prominent topics over from 2010 to 2022 against the background of South Korea’s main socioeconomic issues. As shown in Fig. 3 and Table 3, “[T6] Department stores’ broad retail strategies” (21.5%, 21.3%) were prominent from 2010 to 2011, coinciding with retailers integrating information technologies into their retail strategies in South Korea (Im et al., 2010; Jo, 2011). Furthermore, the order of the predominant keywords “profit” and “performance” interchanged.

In 2012, “[T8] Sale of distressed real estate assets” (15.8%) prevailed, accompanied by the alternating keywords “building” and “development.” At the same time, Samsung

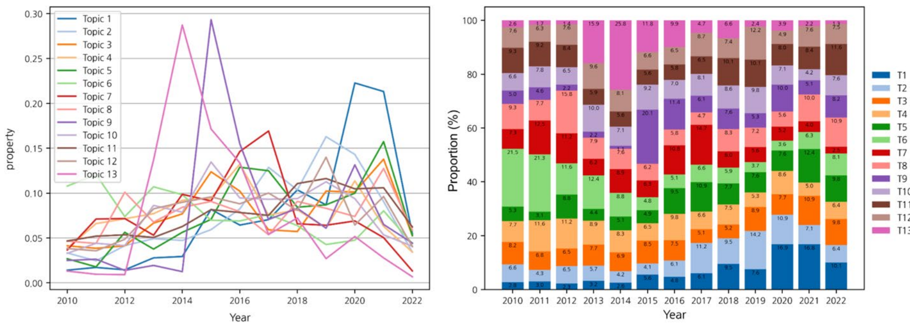


Fig. 3 Results of time series topic analysis and property changes

Table 3 Comparing South Korea's socioeconomic issues and topic changes in primary insolvent K-Fashion enterprises (2010–2022)

Year	The socioeconomic issues in South Korea	The dominant topic of news about insolvent K-Fashion enterprises in DTM results	The top five keywords' proportion changes	
2010	Beginning the 'Tablet PC Era' and overheating competition in the retail industry (Im et al., 2010)	[T6] Department stores' broad retail strategies (21.5%, 21.3%)	[2010] improvement turnover **profit** **performance** sales	[2011] improvement turnover **performance** **profit** sales
2011	The emergence of new industries due to the spread and convergence of IT technology (Jo, 2011)			
2012	Concerns about the Korean economy's triple crisis: low growth, high prices, household debt, Park Geun-hye elected president (Samsung Global Research, 2012)	[T8] Sale of distressed real estate assets (15.8%)	[2011] real estate Dongdaemun **building** **development** Seoul	[2012] real estate Dongdaemun **development** **building** Seoul
2013	Current account and recession-type surplus (Chosun Biz, 2013)	[T13] Business diversification through new affiliates (15.9%, 25.8%)	[2013] merger trading firm **manpower** **construction firm** integration	[2014] merger trading firm **construction firm** **manpower** integration
2014	The opening and influence of the "Korean Wave (*Hallyu*)" in the manufacturing industry (Seoul Institute, 2014)			
2015	Prolonged low growth and low prices due to continued deflation concerns (Han, 2015)	[T9] Court receivership of sports outdoor brands (20.1%, 11.4%)	[2015] stagnation depression **outdoor** **sports** FILA	[2016] stagnation depression **sports** **outdoor** FILA
2016	Economic recession with 0% growth for four-consecutive quarters and the impeachment of President Park Geun-hye (Chosun Biz, 2016)			
2017	Brexit-induced economic uncertainty and rapid regulatory reform for business investment, President Moon Jae-in elected (Korea Federation of SMEs, 2017)	[T7] Samsung C&T's corporate governance changes (14.7%)	[2017] Samsung C&T president equity Samsung Group Cheil Industries	[2018] Samsung C&T president equity Samsung Group Cheil Industries
2018	Fostering future industries based on income-led growth and the 4th industrial revolution (KPMG, 2018)			
2019	Structural long-term slump, consumption imbalance, domestic impact of U.S. economic recovery (Hong & Oh, 2018)	[T11] Domestic performance of global fast fashion brands (10.1%)	[2018] success **USA** **Japan** UNIQLO fast fashion	[2019] success **Japan** **USA** UNIQLO fast fashion
2020	COVID-19 pandemic hits face-to-face services (Statistics Korea, 2020), competition in the 4th industry in advanced countries intensifies (Bank of Korea, 2020)	[T1] Performance improvement policy (16.9%, 16.8%)	[2020] turnover **sales** **profit** performance improvement	[2021] turnover **profit** **sales** performance improvement
2021	Damage to the global supply chain, continued inflation, accelerating carbon neutrality, Upward surge in real estate price (The Federation of Korean Industries, 2021)			
2022	Rising interest rates, year-round high inflation, trade deficit, President Yoon Seok-yeol elected (Yoo & Ahn, 2022)	[T11] Domestic performance of global fast fashion brands (11.6%)	[2021] **Japan** **USA** **success** **UNIQLO** fast fashion	[2022] **USA** **Japan** **UNIQLO** **success** fast fashion

Note: 1) [T1]–[T13] refer to Topics 1 through 13; only selected topics are presented in this table.
2) DTM categories

Global Research (2012) described South Korea's economic challenges as its low growth, rising inflation, and increasing household debt burdens.

During 2013–2014, "[T13] Business diversification through new affiliates" (15.9%, 25.8%) was dominant, with the keywords "labor" and "construction" varying in significance. In this period, Chosun Biz (2013) discussed concerns about a potential economic downturn, while the Seoul Institute (2014) highlighted *Hallyu*'s role in fostering diversification in the economy.

In 2015 and 2016, "[T9] Court receivership of sports outdoor brands" (20.1%, 11.4%) prevailed, with the keywords "outdoor" and "sports" prominent. The persistent concern about prolonged low economic growth then suggests that this circumstance impacted local outdoor sports brands (Chosun Biz, 2016; Han, 2015).

During 2017–2018, "[T7] Samsung C&T's corporate governance changes" (14.7%) prevailed, with the significance of the major keywords remaining consistent. In these years, the Korea Federation of SMEs (2017) discussed the repercussions of the Brexit-induced impact on the local economy, while KPMG (2018) elucidated the importance of industries associated with the Fourth Industrial Revolution.

In 2019, "[T11] Domestic performance of global fast fashion brands" (10.1%) was significant, with the prominence of the keywords "USA" and "Japan" interchanging. During this period, South Korea endured a prolonged economic downturn, which was influenced by the economic recovery in the United States (Hong & Oh, 2018).

In 2020–2021, "[T1] Performance improvement policy" (16.9%, 16.8%) prevailed, with a shifting focus on the keywords "sales" and "profit." Socioeconomic experts (Bank of Korea, 2020; Statistics Korea, 2020; The Federation of Korean Industries, 2021) noted the multifaceted impact of external factors such as COVID-19, global supply chain disruptions, and surging real estate prices.

In 2022, "[T11] Domestic performance of global fast fashion brands" (11.6%) was prominent, with a focus on the keywords "Japan," "USA," "success," and "UNIQLO." At this time, Yoo and Ahn (2022) underscored the fierce competition among global fashion brands in South Korea due to the impact of the trade deficit.

Discussion

To answer RQ1, Study 1 identified three categories of sequential changes across thirteen topics: strategies for improving sales performance, organizational restructuring, and the influence of global fashion labels. Additionally, to answer RQ2, it confirmed that the time series changes in topic percentages from 2010 to 2022 reflected South Korea's socioeconomic issues. Key topics included department stores' retail strategies 2010–2011, real estate asset sales in 2012, and business diversification through new affiliates 2013–2014. In the years 2015–2016, sports brand receivership was prominent, followed by corporate governance changes at Samsung C&T 2017–2018. From 2019 onwards, significant topics were regarding the performance of global fast fashion brands and policies for performance improvement. This reflected the impacts of economic downturns, the Fourth Industrial Revolution, and COVID-19 on the industry.

Study 2

In Study 2, we aggregated KOSPI data for Textile & Wearing Apparel and macroeconomic indices, aligning them with Study 1's topics and calculating weekly averages. After validating the descriptive statistics, we used multiple linear regression analysis and diagnostic procedures to identify which topics of insolvent K-Fashion firms significantly impacted stock market indices, indicating RQ3 and RQ3.

Table 4 Summary of the variable's introduction and data collection

Type	Variables	Definition	Raw data source
Dependent variable	Clothing firms in the KOSPI	the index of textile and clothing industry stocks traded on the stock market division in South Korea	Korea Exchange (http://www.krx.co.kr)
Independent variable (Industrial factors)	The insolvent K-Fashion firms news news topic index	Topic percent contribution*weights index derived from latent Dirichlet allocation algorithm	NAVER online news (https://news.naver.com)
Independent variable (Economy factors)	Employment growth rate change	The percentage change of the employed among the population aged 15 and over	Bank of Korea (http://bok.or.kr)
	GDP growth rate change	The percentage change in the sum of the market values of goods and services produced in a country	
	Housing transaction growth rate	The percent change in the price index of the actual amount exchanged in a housing contract	

Methods

Data collection

For the clothing firms in the KOSPI (Kim & Park, 2023), we collected data on the date, closing price, growth rate, and market price from the Korea Exchange for 2010–2022. Additionally, we gathered economy-related independent variables such as the employment rate (Latif et al., 2021), GDP growth rate (Mukherjee et al., 2023), and housing transaction growth rate (Jiao et al., 2022) from the Bank of Korea. Furthermore, the news topic data (e.g., topic number, date, and topic index) for insolvent K-Fashion enterprises used in Study 1 were adopted. Table 4 shows the definitions of the variables and data sources.

Refining and cleansing the data

The data on the topic variables were structured and stored according to the topic number, date, and topic index values, while only the date, closing price, and rate of change from the KOSPI data were retained. Then, we reformatted the macroeconomic data, followed by addressing the topic variables. All the symbols embedded in the different date formats were standardized. Consequently, we created a SAS dataset unifying each variable based on the topic's weekly data using the univariate function, which computes the weekly average considering the number of weekly dates. Missing values for topics and days without stock trading were replaced with zero.

Measurement

We analyzed the descriptive statistics to verify the overall data volume and various statistical measures. Subsequently, we performed multiple regression analysis, incorporating the parameter estimates, standard errors, t-values, p-values, and variance inflation factors. We checked the model's fit and diagnosis plot and then formulated the regression equation to examine the impact of the industrial and macroeconomic variables on

Table 5 Descriptive statistics

Variables	N	Mean	S.D	Min	Max	Median	Variance	Skewness	Kurtosis
Average textile & wearing Apparel KOSPI	668	286.11	**89.94**	0	542.59	278.20	8088.54	**0.30**	0.89
Average [T1]	668	1.30	1.27	0	10.10	1	1.61	2.19	7.36
Average [T2]	668	1.47	1.95	0	28.74	0.99	3.8	5.89	64.70
Average [T3]	668	0.62	0.95	0	8.16	0.33	0.91	3.52	16.93
Average [T4]	668	1.63	1.33	0	8.09	1.3	1.76	1.74	4.09
Average [T5]	668	2.44	2.7	0	27.57	1.72	7.31	3.62	21.06
Average [T6]	668	3.19	3.64	0	38.29	2.27	13.29	3.56	20.27
Average [T7]	668	1.57	1.92	0	12.42	1	3.67	2.29	6.84
Average [T8]	668	1.95	1.97	0	27.38	1.51	3.87	4.40	43.91
Average [T9]	668	0.35	0.65	0	5.59	0.07	0.42	3.60	17.64
Average [T10]	668	1.28	1.14	0	7.29	0.98	1.31	1.91	5.07
Average [T11]	668	2.25	1.82	0	13.70	1.82	3.33	1.90	5.76
Average [T12]	668	1.75	2.12	0	23.13	1.18	4.51	4.05	26.85
Average [T13]	668	2.31	3.32	0	31.61	1.39	11.01	3.65	18.49
Employment rate change	668	60.26	0.80	**58.30**	62.3	60.5	0.64	**− 0.15**	− 0.05
GDP change rate	668	3.06	1.75	**− 2.50**	7.9	2.90	3.05	**− 0.04**	2.62
Housing transaction change rate	668	0.3	0.38	**− 0.20**	1.5	0.2	0.14	1.57	1.87

[T1]–[T13]: Topics 1 to 13

Bold values indicate key statistical measures relevant to the analysis

the clothing firms in the KOSPI. To prevent human error during the process and analysis, all the measurements were meticulously taken using the SAS program.

Study 2 results

Descriptive statistics

After a preprocessing procedure involving the calculation of weekly averages based on each topic, the dataset contained 668 entries (Table 5). The KOSPI measures showed rising values compared with the other variables due to the closing price, resulting in a high standard deviation (89.94) and skewness of 0.30. These findings suggest a pronounced fluctuation in the KOSPI. Additionally, stock market holidays are shown by a minimum average KOSPI value of zero. A zero for the minimum value of the topic variable indicates that no news relevant to the specified topic was released during that period. In addition, the minimum values of the macroeconomic variable were negative (i.e., a GDP growth rate of − 2.5 and a housing transaction growth rate of − 0.20), while the employment growth rate was also low (58.30). In terms of skewness, KOSPI (0.30), the employment growth rate (− 0.15), and the GDP growth rate (− 0.04) demonstrated the data's normal distribution given that the values were close to zero.

Multiple linear regression analysis and model diagnosis

We employed multiple regression analysis to measure the effect of news on insolvent K-Fashion firms and the macroeconomic variables on the clothing firms in the KOSPI (Table 6). The coefficient values of four topics were significant: "[T3] Management

Table 6 Results of multiple regression analysis

Variable	Parameter estimate	Standard error	*t*-Value	Variance inflation
Intercept	− 4174.16	194.08	− 21.51***	0
[T3] Management control through equity disclosure and acquisition	7.32	2.53	**2.89****	1.01
[T5] Bankruptcy and delisting of local K-Fashion brands	2.08	0.92	**2.27***	1.06
[T7] Samsung C&T's corporate governance changes	3.70	1.28	**2.89****	1.04
[T13] Business diversification through new affiliates	1.98	0.73	**2.73****	1.00
Employment growth rate change	73.05	3.21	**22.77*****	1.15
GDP growth rate change	5.03	1.43	**3.52****	1.07
Housing transaction growth rate	75.40	6.40	**11.78*****	1.01
Root MSE = 62.09, R-Square = .53, Adj R-Sq = .52				

[T1]–[T13] refer to Topics 1 through 13; only selected topics are presented in this table

* $p < .05$, ** $p < .01$, *** $p < .001$

control through equity disclosure and acquisition" ($\beta = 7.32$; $SE = 2.53$; t-value = 2.89; $p = 0.00$), "[T5] Bankruptcy and delisting of local K-Fashion brands" ($\beta = 2.08$; $SE = 0.92$; t-value = 2.27; $p = 0.02$), "[T7] Samsung C&T's corporate governance changes" ($\beta = 3.70$; $SE = 1.28$; t-value = 2.89; $p = 0.00$), and "[T13] Business diversification through new affiliates" ($\beta = 1.98$; $SE = 0.73$; t-value = 2.73; $p = 0.00$). Additionally, the coefficients of three macroeconomic variables were significant: employment growth rate change ($\beta = 73.05$; $SE = 3.21$; t-value = 22.77; $p < 0.0001$), GDP growth rate change ($\beta = 5.03$; $SE = 1.43$; t-value = 3.52; $p = 0.00$), and housing transaction growth rate ($\beta = 75.40$; $SE = 6.40$; t-value = 11.78; $p < 0.0001$). The variation inflation factors ranged from 1.00 to 1.15, namely, below the recommended threshold of 5, which suggested a good fit. Furthermore, the F value (105.61) was significant ($p < 0.0001$) and 53% of the variation in the KOSPI was explained by the inclusion of the industrial (e.g., [T3], [T5], [T7], [T13]) and macroeconomic variables (e.g., Employment growth rate change, GDP growth rate change, housing transaction growth rate) (Root MSE = 62.09; R-Square = 0.53; Adj R-Sq = 0.52), indicating a moderate level of correlation. Accordingly, the multiple linear regression model is presented in Eq. (2).

The regression equation of K-Fashion's insolvent enterprise news and macro-economic impact on clothing industry's business cycle

$$\begin{aligned}\widehat{Y} = &- 4{,}174.156 + 7.316X1 + 2.082X2 + 3.695X3 + 1.980X4 \\ &+ 73.054X5 + 5.026X6 + 75.404X7 + \varepsilon\end{aligned} \tag{2}$$

$\widehat{Y}$: Textile & Wearing Apparel KOSPI; $X1$: [T3] Management control through equity disclosure and acquisition; $X2$: [T5] Bankruptcy and delisting of local K-Fashion brands; $X3$: [T7] Samsung C&T's corporate governance changes; $X4_{t13}$: [T13] Business diversification via new affiliates; $X5$: Employment growth rate change $X6$: GDP growth rate change $X7$: Housing transaction change rate. Note: [T1]–[T13] refer to Topics 1 through 13; only selected topics are presented in this table.

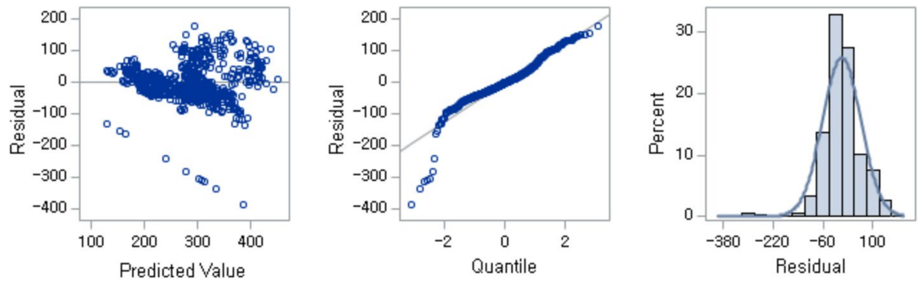


Fig. 4 Fit diagnostics for the multiple regression model

To assess the goodness of the fit of the multiple regression model, we examined the raw residuals (Fig. 4). These exhibited no discernible pattern, suggesting a random distribution of the variables. The quantile–quantile plot indicated that the residuals closely approximated the normal distribution along the diagonal line.

Discussion

To answer RQ3, Study 2 revealed that the business cycle of the fashion industry was notably affected by equity reporting, financial collapse, Samsung C&T's corporate governance reforms, and the diversification of business operations. Additionally, macroeconomic variations in the employment rate, GDP growth, and housing transaction growth rate had a positive impact, as found in RQ4.

Discussion and Conclusions

Theoretical contributions

Drawing on CLC theory (Hasan & Habib, 2017; Zhang et al., 2021), this study empirically validated the proposed classification of online news on insolvent K-Fashion firms over from 2010 to 2022 into sales performance improvement strategies (e.g., [T1], [T3], [T4], [T6], [T10], [T13]), corporate restructuring and financial challenges (e.g., [T2], [T5], [T7], [T8], [T9], [T12]), and the local market influence of global fashion brands (e.g., [T11]). Numerous local fashion firms have encountered reduced sales since the early 2010s (Korea Federation of SMEs, 2016; Ryu, 2011). Therefore, our results chronologically indicated the evolution of media topics over the past twelve years through data mining analysis.

Additionally, the present study is novel methodologically, as it integrates various approaches to big data analysis, encompassing text mining techniques such as DTM, along with structured data derived from the clothing firms in the KOSPI. Earlier studies have examined the qualitative aspects of fashion firm's life cycles (Yu, 2021), conducted time series regression analysis of news on the fashion industry (Kim & Park, 2023), and adopted machine learning models to assess clothing sales (Lv et al., 2023). This study's enhanced approach extends these investigations by incorporating both natural language processing and structured data.

Consistent with prior research elucidating the characteristics of the decline stage of the CLC model (Hasan & Habib, 2017; Zhang et al., 2021), which is marked by diminished profits and organizational restructuring (Faff et al., 2016; Filatotchev et al., 2006; Wen et al., 2022), this study's findings delineated a pattern specific to insolvent K-Fashion firms. In parallel, Yu, (2021) clarified the business cycle of South Korea's fashion industry using qualitative methods and Ryu (2011) documented the substantial rise in competitive pressure in the K-Fashion sector driven by factors such as the intervention of the IMF, the rapid expansion of global fashion brands, and global financial crisis during the 2010s. Moreover, Cho and Chyung (2020) found that the COVID-19 pandemic affected the K-Fashion industry by interrupting production processes and causing a significant drop in retail sales. Building on this body of research, this study found noticeable shifts in news topics on insolvent K-Fashion firms. This was obtained by employing comprehensive big data analysis, including a media news assessment.

Furthermore, this study examined the socioeconomic issues of South Korea during the decline stage of the K-Fashion industry. This was achieved through a comparative analysis of prominent topics and keyword changes. Previous research has highlighted the primary societal challenges that collectively contributed to the economic recession in the early 2010s. It includes the country's low growth rate, rising inflation, and burden of household debt (Han, 2015; Im et al., 2010; Jo, 2011; Samsung Global Research, 2012; Seoul Institute, 2014). We find that 2010–2014 online news on insolvent K-Fashion firms predominantly emphasized strategies to increase sales, the establishment of new affiliates through diversification, and real estate transactions involving distressed assets. We followed prior research highlighting the ramifications on the local economy arising from Brexit, advancements owing to the development of the Fourth Industrial Revolution, and prolonged subdued economic growth (Korea Federation of SMEs, 2016; KPMG, 2018) . By doing so, we found that in the mid-2010s, there were instances of court receiverships involving local fashion sportswear brands and corporate governance restructuring within Samsung C&T. We found that 2016–2019 factors that significantly shaped the industry, included imbalanced consumption and the local implications of the U.S. economic recovery (Hong & Oh, 2018). This was followed by the outbreaks of COVID-19 in late 2019, which reshaped the fashion industry attributed to the global pandemic (Bank of Korea, 2020). Hence, our study showed that the insolvency issues among K-Fashion firms were linked to policies aimed at increasing sales and to the intense competition from global fast fashion brands.

Following the effects of media priming theory (Perloff, 2022; Piselli et al., 2022), we further explored the relationship between the online news coverage of financially distressed K-Fashion firms and the stock market indices of clothing companies. Concurring with Hoewe (2020) and Moy et al. (2016), we found that the media priming effect is triggered when individuals recall pertinent information from media news. This subsequently influenced their behavioral intention. Findings indicated that online news topics related to insolvent K-Fashion firms influenced the clothing industry's stock market index. This particularly concerned management's decisions on equity disclosure and acquisition, the delisting of local fashion brands, Samsung C&T's governance changes, and diversification by creating new affiliates. Other scholars have investigated the impact of economic news on the stock market (Nam & Seong, 2019), the public's perceptions of the government (Kalogeropoulos et al., 2017), and oil market fluctuations (Mensi et al., 2014) within the framework of the news priming effect. Our study is consistent with previous research that has found a robust correlation between corporate news and the stock market impact (Clarke et al., 2020; Hanna et al., 2020; Lee & Brahmasrene, 2018; Shynkevich et al., 2016). An additional noteworthy contribution of our study is its examination of both unstructured data (e.g., news on the insolvency of K-Fashion firms) and financial data (e.g., the clothing firms in the KOSPI).

This study also provides theoretical insights, indicating that changes in the employment rate and housing transaction growth rate positively influence the business cycle of the clothing industry. This impact extends beyond the scope of news content on insolvent K-Fashion firms. As highlighted by Holmes and Maghrebi (2016) and Li et al. (2015), employment and housing transaction growth rates are significant indicators for gauging a country's social well-being and economic health. Recent research has assessed economic indicators across various sectors such as the metals industry (Dinh et al., 2022), oil and gas industry (Morgunova & Shaton, 2022), and banking industry (O'Connell, 2023). However, limited empirical evidence is available on the decline stage of K-Fashion firms. Given the multifaceted nature of the fashion industry, which encompasses diverse business divisions such as production, distribution, and retail (Kim & Park, 2023), this study bridges the existing knowledge gap by shedding light on the pivotal macroeconomic influences shaping the K-Fashion industry.

Managerial implications

The study's findings illuminate the changing trends in online news topics and keywords related to insolvent firms in the K-Fashion sector over 2010–2022. These insights can be a valuable resource for South Korean government agencies seeking to formulate support policies for the K-Fashion industry. With issues such as reduced demand caused by the global economic slowdown, sluggish exports, and rising energy costs (Ministry of Trade, Industry & Energy, 2023), the government has allocated 17.4 billion won to address these challenges (Lee, 2023). The findings of this study highlight the urgency of these efforts, as evidenced by a recent survey that found a 27% closure rate among online clothing retailers within three years of establishment (Shin, 2022). By utilizing the trends and insolvency categorizations identified in this research, policymakers can implement more nuanced, data-driven interventions to support struggling K-Fashion firms.

For K-Fashion company owners and managers, the findings offer practical insights into identifying financial risks during economic downturn. The categorizations of insolvency-related news topics can serve as tools for managers to assess the status of their companies and anticipate potential future risks. These insights can inform strategic decision-making, such as restructuring initiatives or resource allocation, aimed at mitigating potential losses. Additionally, foreign brands aiming to penetrate the K-Fashion market can use these findings to better understand market volatility and develop strategies to navigate this competitive and evolving landscape.

The research also has significant implications for international stakeholders. For foreign brands and international investors, understanding the relationship between news narratives and market performance provides a strategic advantage when evaluating opportunities in South Korea's fashion industry. By analyzing the trends identified in this study, international investors and brands can gain a deeper understanding of market risks, consumer behavior, and broader economic factors that influence the K-Fashion industry.

Finally, investors in the fashion-related stock market could acquire valuable insights for minimizing their risk exposure, particularly on those news topics that influence the fashion business cycle. The empirical findings of this study highlight the correlation between the news coverage of the insolvency of K-Fashion firms and stock market performance in the clothing industry, offering investors foresight to mitigate potential losses. Individual investors could juxtapose the latest news on fashion firms against the insolvency categorizations delineated in this study.

Limitations and suggestions for future research

This study had several limitations. Study 1 was restricted to 2010–2022, the decline stage of the business cycle. While many scholars have used similar timeframes to capture trends (Daly, 2022; Meng et al., 2022), such an approach may inadequately depict the entire CLC encompassing the inception, growth, and shake-out stages. Consequently, future research could examine the complete CLC from the 1960s to better understand the formation of K-Fashion enterprises.

The second limitation was the exclusion of qualitative information on K-Fashion firms in Study 2. To comprehend the distinct attributes of the owners of K-Fashion firms, it would be beneficial to gather information on the composition of the board of directors and assess executives' competencies to manage these organizations effectively. Additionally, conducting focus group interviews with owners and assessing individual corporate risk factors could offer valuable insights for examining the decline stages of diverse fashion businesses (e.g., textile, manufacturing, and retailing), along with their consequential effects on the stock market.

Third, this study utilized South Korea's largest online portal site's media affiliates to gather digital news on K-Fashion firms' insolvencies. Despite its popularity and the enormous amount of available information, it is limited in capturing the global impacts of the international economy and cross-societal factors on K-Fashion firms. Since the fashion business cycles are widely affected by global trade and distribution within the overseas economy, future studies may consider including global Korean news sources that potentially influence Korea's stock market index in the fashion industry.

Finally, this study found the dominance of macroeconomic variables largely influenced the fashion industry's business cycle, while the failing fashion firms had smaller impacts. However, the results of industry-specific topics on the decline of fashion firms derived from this study provide crucial insights into how particular issues and decisions within failing fashion firms contribute to broader industry dynamics. Future studies should enhance these relationships, integrating the intrinsic components of the fashion value chain, including clothing balance trade (Rahman et al., 2019) and garment production level (Chen et al., 2021), to contribute to broader industry dynamics within the fashion sector.

Abbreviations

CLC	Corporate life cycle
DTM	Dynamic topic modeling
IMF	International Monetary Fund

LDA Latent Dirichlet allocation
MFA Multi fibre arrangement
TF Term frequency
TF-IDF Term frequency-inverse document frequency

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Hyojung Kim: Conceptualization, Investigation, Data curation, Formal analysis, Methodology, Project administration, Writing—original draft, Supervision, Supervision, Funding acquisition, Writing—review & editing.

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Availability of data and materials
The datasets used and analyzed during the current study are available from the main author upon reasonable request.

Declarations

Competing interests
The authors declare that they do not have any competing interests.

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